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CLIMATE



How a New Jersey
startup found an
electrifying way to slash
copper costs



Skyrocketing demand for copper promises to push prices to new heights. As the global economy transitions away from fossil fuels, it is going to need twice as much copper in the coming years than humanity has mined throughout all of its existence.

Still Bright, a New Jersey-based startup founded in 2022, thinks it has found a novel (and cleaner way) to slash those costs.

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IMAGE CREDITS: MONTY RAKUSEN / GETTY IMAGES

“The fact that we’ve already mined the easily mineable stuff, and the fact that we need many more mines to come into production every year — we’re talking like 60-plus mines — is it seems like an impossibility, like there’s no path to get there,” Randy Allen, co-founder and CEO of Still Bright, told TechCrunch.

But a large fraction of that demand could be met if companies can extract more copper from the ore they already mine.

Still Bright has developed a new way to extract copper, one that it says can recover nearly all the copper from typical ores without pre-processing steps that lose up to 20% of the metal. It’s effective enough that it could even be used on tailings, the

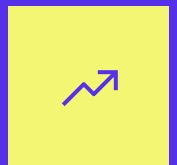
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discard piles that still contain smaller amounts of the metal that mines leave behind.

“Any copper that was lost as waste, we can actually process that and get the copper back,” Allen said.

To boost production from single digits to hundreds of tons per year, Still Bright has raised an \$18.7 million seed round led by Material Impact and Breakthrough Energy Ventures, the company exclusively told TechCrunch. Azolla Ventures, Fortescue, Impact Science Ventures, and SOSV participated.

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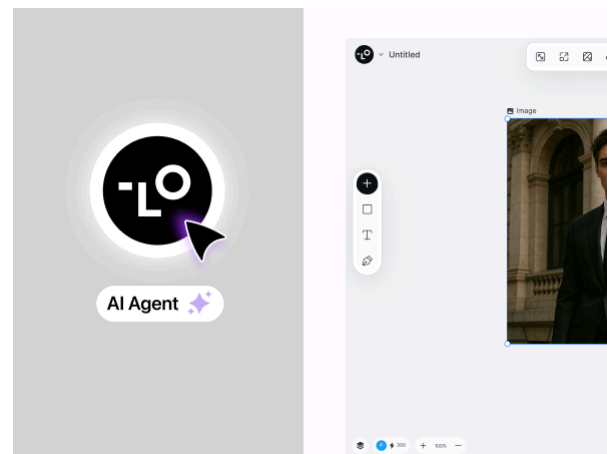
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Still Bright’s technology allows it to extract copper without producing harmful pollution. Where most companies essentially burn away unwanted parts of the ore — releasing much of it into the atmosphere — Still Bright soaks copper-containing ores in a vanadium-based solution, which draws the metal out of the ore. When the vanadium solution is spent, the company’s system uses electricity to regenerate it.

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The core technology was inspired by a type of long-duration energy storage known as a “vanadium flow battery.” In it, a vanadium-based solution that can be stored in large tanks is charged and discharged by flowing it past a membrane.

“All of this is kind of happenstance. The technical inventor, the CTO of the company Jon [Vardner], he was working on two different projects,” Allen said, one on vanadium flow batteries and another on using vanadium to extract copper. “It’s connecting the dots. One person happened to be doing both.”

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Still Bright’s modular system will be able to be installed at mines spanning a range of sizes. Because the vanadium-based process works so quickly, the company’s equipment is much smaller than a typical refiner for the same amount of copper production. “The processing is on the order of minutes, up to an hour. That allows us to keep everything really small.”

The small size pays financial dividends, too. Allen said that Still Bright’s equipment is 70% to 90% cheaper than typical pyrometallurgical gear. Currently, the company’s process costs about the same to run as a typical refinery, but Allen expects that to change. “There’s a lot of opportunity for us to be cheaper,” he said.

Still Bright is planning to build a demonstration unit in 2027 or 2028 that’s capable of producing 500 tons of copper annually. It’s a big leap from the current pilot scale unit, which makes 2 tons per year. The ultimate commercial-scale system will produce 10,000 tons per year.

The clock is ticking, though. Still Bright would like to start refining copper in large enough quantities to benefit from any tariffs President Trump might impose on imports of the metal. If it does, it can use that revenue to develop and deploy commercial-scale units.

“We see ourselves as having a path to be among the cheapest copper producers,” Allen said.

This article has been updated to correct the name of one of the investors. Azolla Ventures, not Apollo Ventures, invested in Still Bright.

Topics: [Climate](#) [copper](#) [Startups](#) [TC](#) [Venture](#)





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Tim De Chant is a senior climate reporter at TechCrunch. He has written for a wide range of publications, including...

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Tesla shuts down Dojo, the AI training supercomputer that Musk said would be key to full self-driving

Rebecca Bellan — 3:19 PM PDT · August 7, 2025

Tesla is breaking up the team behind its [Dojo supercomputer](#), ending the automaker's play at developing in-house chips for driverless technology, according to [Bloomberg](#).

Dojo's lead, Peter Bannon, is leaving the company, and the remaining team members will be reassigned to other data center and compute projects within Tesla, per Bloomberg's reporting, which cited anonymous sources.

The disbanding of Tesla's Dojo efforts follows the departure of around 20 workers, who left the automaker to start their own [AI company called DensityAI](#). The new startup is reportedly coming out of stealth soon and is building chips, hardware, and software that will power data centers for AI that are used in robotics, by AI agents, and in automotive applications. DensityAI was founded by former Dojo head [Ganesh Venkataramanan](#) and ex-Tesla employees Bill Chang and Ben Floering.

It also comes at a crucial time for Tesla.

[CEO Elon Musk has pushed](#) to get shareholders to view Tesla as an AI and robotics company, despite a

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limited robotaxi launch in Austin this past June that featured Model Y vehicles with a human in the front passenger seat and resulted in a number of reported incidents of the vehicles exhibiting problematic driving behavior.

Tesla’s decision to shut down Dojo, which Musk has been talking about since 2019, is a major shift in strategy. Musk has said that Dojo would be the cornerstone of Tesla’s AI ambitions and its goal to reach full self-driving due to its ability to “process truly vast amounts of video data.” He talked about Dojo, albeit briefly, as recently as the company’s second-quarter earnings call.

In 2023, Morgan Stanley predicted Dojo could add \$500 billion to the company’s market value by unlocking new revenue streams in the form of robotaxis and software services. Just last year, Musk noted that Tesla’s AI team would “[double down](#)” on Dojo in the lead-up to [Tesla’s robotaxi reveal](#), which happened in October.

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But talk about Dojo halted around August 2024, when Musk began touting Cortex instead, Tesla’s “giant new AI training supercluster being built at Tesla HQ in Austin to solve real-world AI.”

The Dojo project was one part supercomputer, one part in-house chip-making. Tesla unveiled its D1 chip when it formally announced Dojo at its first AI Day in 2021. [Venkataramanan](#) presented the chip, which Tesla said would be used alongside Nvidia’s GPU to power the Dojo supercomputer. The automaker also said it was working on a next-gen D2 chip that would solve any information flow bottlenecks of its predecessor.

Sources told Bloomberg that now Tesla plans to increase its reliance on Nvidia, as well as other external tech partners like AMD for compute and Samsung for chip manufacturing. Tesla last month signed a [\\$16.5 billion deal with Samsung](#) to make its AI6 inference chips, a chip design that promises to scale from powering FSD and Tesla’s Optimus humanoid robots all the way to high-performance AI training in data centers.

During Tesla’s second-quarter earnings call, Musk hinted at potential redundancies.

“Thinking about Dojo 3 and the AI6 inference chip, it seems like intuitively, we want to try to find convergence there, where it’s basically the same chip,” Musk said.

The news comes as [Tesla’s board offers Musk a \\$29 billion pay package](#) to keep him at Tesla and help push the company’s AI efforts forward, rather than getting too sidetracked by his other companies, including the more pure-play AI startup xAI.

TechCrunch has reached out to Tesla for more information.

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Rocket Lab eyes big defense opportunities with new acquisition

Aria Alamalhodaie — 2:23 PM PDT · August 7, 2025

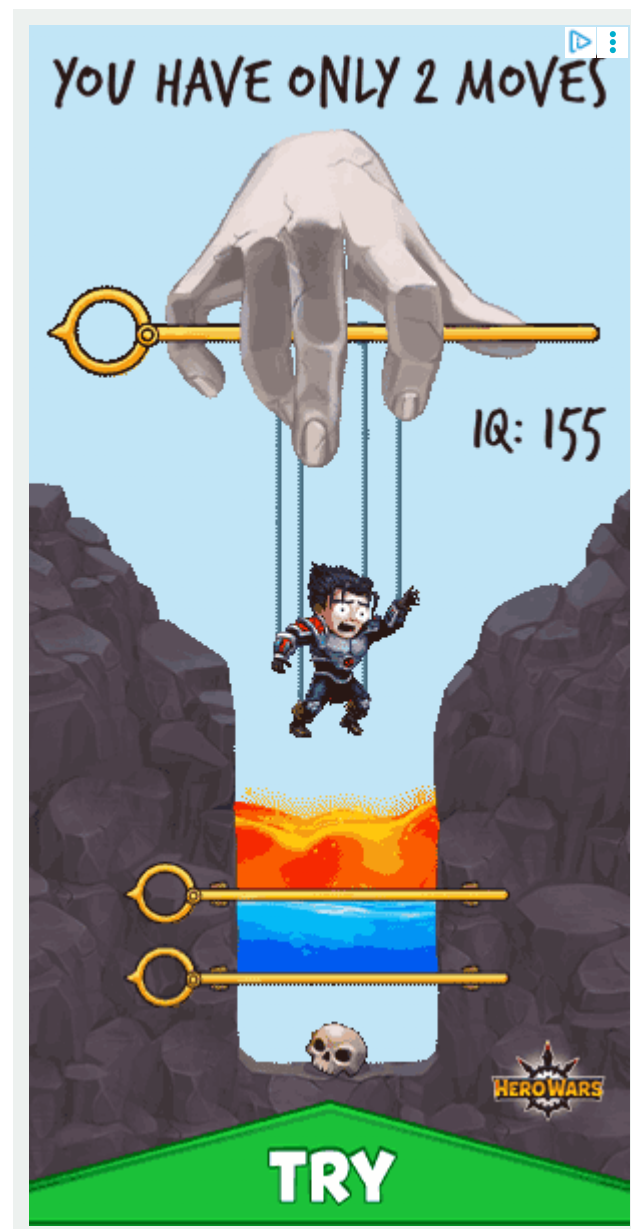
Rocket Lab is signaling to investors, yet again, that it's more than "just" a rocket company.

Rocket Lab's second-quarter results, which were posted Thursday, show revenues continue to be driven by its space systems business rather than launch. The results also highlighted the company's acquisition strategy and how its purchase of a new optical payloads company will make it more competitive for lucrative government contracts.

IMAGE CREDITS: [ROCKET LAB](#)

The company's space systems brought in \$97.9 million of the \$144.5 million in total revenues for the second quarter. Rocket Labs' total revenue, its highest quarterly revenue in the company's history, jumped 36% from a year ago. The company's net loss widened to \$66.4 million.

Rocket Lab founder and CEO Peter Beck highlighted the "busy quarter of M&A activity" as the company gets close to closing its deal to buy [Geost](#), a company that builds optical payloads. Through that acquisition, which will close at \$275 million in cash and equity, Rocket Lab is opening a new business unit called Optical Systems to scale electro-optical and infrared sensor manufacturing.



Latest in Space

Those sensors are used in missile warning, tracking, and space domain awareness. The acquisition is also one part of a larger play from Rocket Lab — spelled out quite explicitly in the company’s earnings presentation — to bid for multibillion-dollar [DOD initiatives like Golden Dome](#).

Rocket Lab has already won some major defense contracts, including a \$515 million, 18-satellite build contract to support the Space Development Agency’s constellation of missile-tracking satellites. The company has moved into production on those satellites, following confirmation that they meet the DOD’s mission requirements, Rocket Lab said.

Down on Earth, Rocket Lab says it’s progressing well toward the first launch of its larger Neutron rocket. The rocket’s launch complex in Virginia is expected to be complete in the third quarter of this year, with [hardware en route](#) and the new rocket engine, called Archimedes, undergoing multiple tests per day.

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Rocket Lab eyes big defense opportunities with new acquisition

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The company is staying vague about the exact launch date, saying only in the presentation that it is pushing an “all-out effort to get Neutron to the launch pad before the end of 2025.”

Rocket Lab ended the quarter with \$564 million in cash and cash equivalents and estimates revenues will hit between \$145 to \$155 million next quarter.

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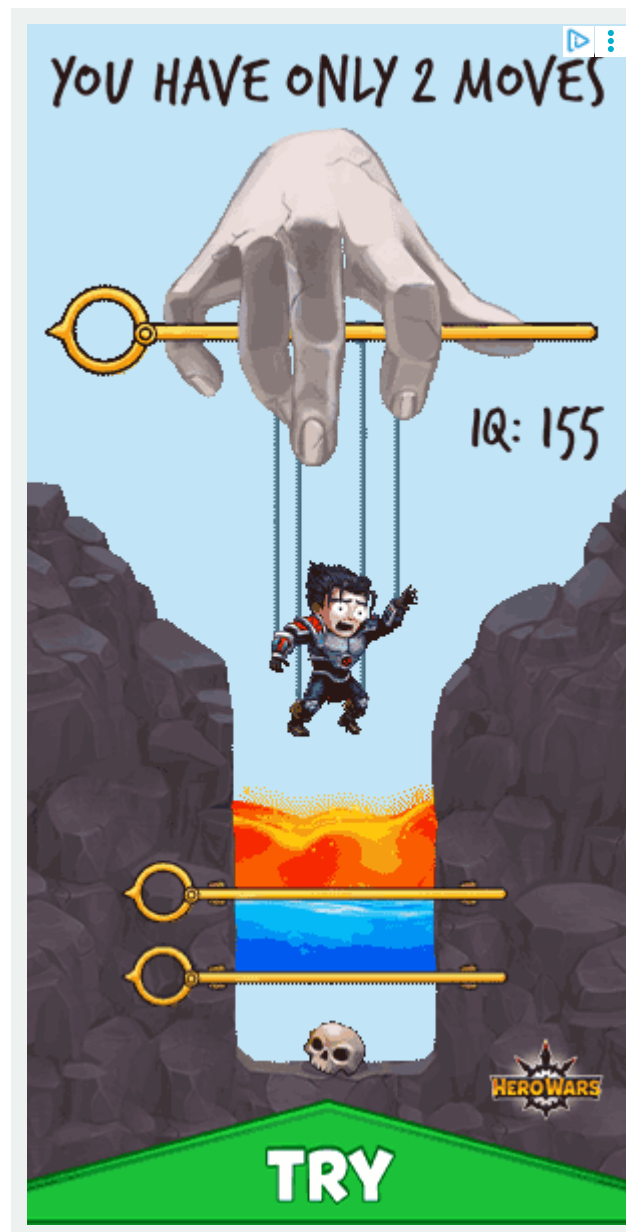


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Aria Alamalhodaie covers the space and defense industries at TechCrunch. Previously, she covered the...

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STARTUPS



High costs and thin margins threatening AI coding startups

Marina Temkin — 2:05 PM PDT · August 7, 2025

In February, AI coding startup Windsurf was in talks to raise a big new round at a [\\$2.85 billion valuation](#) led by Kleiner Perkins. At double the valuation it hit six months earlier, sources told TechCrunch at the time. That deal didn't happen, according to a source familiar with the matter. Instead, news broke in April that the startup planned to sell itself to OpenAI for roughly the same valuation: [\\$3 billion](#).

IMAGE CREDITS:
YUICHIRO CHINO / GETTY IMAGES

While that deal famously fell apart, one bigger question remains: If the startup was growing that fast and attracting VC interest, why would it sell at all?

Insiders tell TechCrunch that for all the popularity and hype around AI coding assistants, they can actually be massively money-losing businesses. Vibe coders generally, and Windsurf in particular, can have such expensive structures that their gross margins are “very negative,” one person close to Windsurf told TechCrunch. Meaning it cost more to run the product than the startup could charge for it.

This is due to the high costs of using large language models (LLMs), the person explained. AI coding

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assistants are particularly pressured to always offer the most recent, most advanced, and most expensive LLMs because model makers are particularly fine-tuning their latest models for improvements in coding and related tasks like debugging.

This is a challenge compounded by fierce competition in the vibe-coding and code-assist market. Rivals include companies that already have huge customer bases like Anysphere's Cursor and GitHub Copilot.

The most straightforward path to improving margins in this business involves the startups building their own models, thereby eliminating costs of paying suppliers like Anthropic and OpenAI.

"It's a very expensive business to run if you're not going to be in the model game," said the person.

- Three weeks after acquiring Windsurf, Cognition offers staff the exit door
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But that idea comes with its own risks. Windsurf's co-founder and CEO, Varun Mohan, ultimately decided against the company building its own model — an expensive undertaking, the person said.

In addition, model makers are already competing directly. Anthropic offers Claude Code and OpenAI offers Codex, for instance.

Selling the business was a strategic move to lock in a high return before it could be undermined by the very companies that supplied its AI, including OpenAI and Anthropic, which were also entering the AI coding market.

Multiple people believe that the same pressure on margins Windsurf faced could be impacting Anysphere, the maker of Cursor, as well as vibe coders like Lovable, Replit, and others.

“Margins on all of the ‘code gen’ products are either neutral or negative. They’re absolutely abysmal,” said Nicholas Charriere, founder of Mocha, a vibe-coding startup and back-end hosting solution serving small and medium businesses (SMBs). He added that he believes the variable costs for all the startups in the sector are very close, likely within 10% to 15% of one another.

Unlike Windsurf, Anysphere has been growing so fast that it intends to remain an independent company, having already [turned down acquisition offers](#), including, reports say, from OpenAI.

And Anysphere [announced in January](#) that it is attempting to build its own model, which could give it more control over its expenses. In July, the startup [hired two leaders from Anthropic’s Claude Code](#) team, the Information reported, but two weeks later, these employees [returned](#) to work at Anthropic.

In addition to building a model, Anysphere could expect the cost of LLMs to decrease over time.

“That’s what everyone’s banking on,” said Erik Nordlander, a general partner at Google Ventures. “The inference cost today, that’s the most expensive it’s ever going to be.”

It’s not entirely clear how true that is. Rather than falling as expected, the [cost of some of the latest AI models has risen](#), as they use more time and computational resources to handle complicated, multistep tasks.

When that will change remains to be seen. On Thursday, for instance, [OpenAI introduced](#) a new flagship model, GPT-5, with fees that are significantly less than its competitor, Anthropic’s Claude Opus 4.1. And Anysphere [immediately offered](#) this model as a choice for Cursor users.

Anysphere has also recently changed its pricing structure to pass along the increased costs of running Anthropic’s latest Claude model, particularly to its most active users. The move caught some of [Cursor customers by surprise](#), since they didn’t expect additional charges on top of its \$20-per-month Pro plan. Anysphere CEO Michael Truell later apologized for unclear communication about the pricing change in a [blog post](#).

This is the rock and the hard place. Although Cursor is one of the most popular AI applications, having reached [\\$500 million in ARR in June](#), the company’s user base may not be so loyal to the product if another company develops a tool that is superior to Cursor, investors say.

Anysphere didn’t respond to a request for comment.

Given the competitive landscape and the costs, Windsurf’s decision to get out may prove to be understandable. After the OpenAI deal fell through, the founders and key employees left to join Google in a deal that led to a [\\$2.4 billion payout](#) to key shareholders. The remaining business then [sold itself to Cognition](#).

While many, including prominent VCs, criticized Mohan for leaving approximately 200 employees without roles at Google, a source familiar with the deal insisted the acquisition actually maximized the outcomes for all employees.

Beyond Cursor, other AI coding tools are also among the fastest growing startups of the LLM generation, like Replit, Lovable, and Bolt, and all of them rely on model makers as well.

Additionally, if this extremely popular business sector, already generating hundreds of millions in revenue or more a year, has difficulty building on top of model makers, what might it mean for other, more nascent industries?

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